

EXAMINATION INVIGILATION DUTY ALLOCATOR

Smart Web-Based System for Automated Duty Management

Software Requirements Document

Course: 25CS1201E — Frontend Development Frameworks & UI Engineering
Academic Year 2025–26 | Term III

Role	Member (Roll No.)
Team Lead	Lahari (2520090226)
Team Member	Bhanu Priya (2520090171)

Table of Contents

- 1. Purpose3
- 2. Problem Statement3
- 3. Goals3
- 4. Scope.....3
 - 4.1 In Scope 4
 - 4.2 Out of Scope..... 4
- 5. Users4
- 6. Functional Requirements.....4
- 7. Non-Functional Requirements5
- 8. Assumptions5
- 9. Acceptance Criteria5

1. Purpose

The purpose of this project is to design a smart, web-based system that automates the allocation of examination invigilation duties. The system ensures the fair, conflict-free and transparent assignment of faculty members across examination halls, while greatly reducing the manual administrative effort traditionally required for this task.

By centralising faculty data, exam schedules and room information on a single platform, the Examination Invigilation Duty Allocator replaces slow, paper-based processes with intelligent, rule-based automation. It delivers balanced workloads, real-time visibility and timely notifications to every faculty member and coordinator involved in the examination process.

2. Problem Statement

In most academic institutions, invigilation duties are still assigned manually using spreadsheets and paperwork. This traditional approach suffers from several recurring problems:

- It is slow and time-consuming, often requiring hours of administrative work for a single exam schedule.
- It is error-prone, leading to mistakes such as double-bookings and missed assignments.
- It results in uneven workload distribution, where some faculty are over-burdened while others are under-utilised.
- It causes scheduling clashes and last-minute confusion among faculty and staff.

These issues reduce efficiency, create dissatisfaction among staff, and make the overall examination process difficult to coordinate. A centralised, automated solution is needed to address them.

3. Goals

The system is built to achieve the following target outcomes:

Goal	Description
Automation	Replace manual allocation with rule-based, automatic duty assignment.
Fairness	Balance the invigilation load evenly across all eligible faculty.
Conflict-Free	Prevent overlaps, double-bookings and schedule clashes.
Transparency	Give administrators and faculty real-time visibility of all duties.
Speed	Cut allocation time from hours of work down to minutes.
Centralization	Keep schedules, rooms and duty records in one single place.

4. Scope

This section defines the boundaries of the system — the features it will deliver and those that are explicitly excluded.

4.1 In Scope

- ✓ Faculty profile and availability management.
- ✓ Automated invigilation duty allocation.
- ✓ Exam scheduling and room / hall allocation.
- ✓ Conflict detection and resolution.
- ✓ Faculty dashboard and notifications.
- ✓ Duty charts and report generation.

4.2 Out of Scope

- ✗ Student exam registration and enrollment.
- ✗ Question paper setting or evaluation.
- ✗ Salary, payroll or HR management.
- ✗ Biometric / CCTV exam monitoring.
- ✗ Native mobile application (the system is web-based only).
- ✗ Integration with external university ERP systems.

5. Users

Three categories of users interact with the system, each with distinct responsibilities:

User	Responsibilities
Administrator	Manages faculty data, configures fairness rules, runs the allocation, approves duties and generates reports.
Faculty / Invigilator	Views assigned duties, dates, timings and hall details; receives notifications and checks duty history.
Exam Coordinator	Sets up exam schedules and room allocations, and monitors conflicts and overall duty status.

6. Functional Requirements

These define what the system must do.

ID	Requirement	Description
FR1	Faculty Management	Add, update and manage faculty profiles, departments and availability.
FR2	Duty Allocation	Auto-assign invigilation duties using availability and fairness rules.

ID	Requirement	Description
FR3	Exam & Room Scheduling	Create exam timetables and allocate examination halls and rooms.
FR4	Conflict Detection	Identify and flag clashes, overlaps and duplicate assignments.
FR5	Notifications	Send real-time alerts to faculty for new or changed duties.
FR6	Reports & Dashboards	Generate duty charts and provide role-based dashboards.

7. Non-Functional Requirements

These define how well the system must perform.

Attribute	Requirement
Performance	Allocate duties for a full exam schedule within a few seconds.
Usability	Clean, intuitive web interface usable with minimal training.
Security	Credential-based login with role-based access control.
Reliability	Consistent, accurate allocation with no loss of data.
Scalability	Support growing numbers of faculty, exams and rooms.
Availability	Accessible anytime through any modern web browser.

8. Assumptions

The system is designed and operated under the following assumptions:

- Faculty availability data is kept up to date.
- Exam schedules and room lists are provided.
- Users have a device with a modern web browser.
- A stable internet connection is available.
- Faculty respond to notifications in good time.

9. Acceptance Criteria

The system will be considered successful when it meets the following measures:

- ✓ Duties are allocated with no scheduling conflicts.
- ✓ Workload is distributed fairly across faculty.
- ✓ Faculty receive accurate, timely notifications.

- ✓ Admin can generate correct duty charts and reports.
- ✓ System runs reliably on standard web browsers.